Т6

а) $L(\theta) = \prod_{i=1}^{n} \frac{\theta-1}{x_i^{\theta}} = \frac{(\theta-1)^n}{(\prod x_i)^{\theta}}$

$\ln L(\theta) = n\ln(\theta-1) - \theta\sum_{i=1}^{n} \ln x_i$

$\frac{d}{d\theta} \ln L = \frac{n}{\theta-1} - \sum \ln x_i = 0$

$\frac{n}{\theta-1} = \sum \ln x_i \Rightarrow \theta - 1 = \frac{n}{\sum \ln x_i}$

$\hat{\theta} = 1 + \frac{1}{\ln x}$

б) проверим, что ВМ регулярна

- $p(x,\theta)$ - непр дифф по θ на $(1,+\infty)$
- $\int_1^{+\infty} \frac{\partial}{\partial\theta} p(x,\theta)\,dx$

$\frac{\partial p}{\partial\theta} = \frac{\partial}{\partial\theta}\left[(\theta-1)x^{-\theta}\right] = x^{-\theta} - \ln x(\theta-1)x^{-\theta} =$

$= \frac{1-(\theta-1)\ln x}{x^{\theta}}$

$\int_1^{+\infty} \frac{\partial p}{\partial\theta}\,dx = \int_1^{+\infty} x^{-\theta}\,dx - (\theta-1)\int_1^{+\infty} x^{-\theta}\ln x\,dx$

$\cancel{\int_1^{+\infty} x^{-\theta}\,dx = \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} = \frac{1}{\theta-1}}$

$$\int_1^{+\infty} x^{-\theta}\,dx = \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} = \frac{1}{\theta-1}$$

$$\int_1^{+\infty} x^{-\theta}\ln x\,dx = \frac{1}{(\theta-1)^2}$$

$$\int_1^{+\infty} \frac{\partial p}{\partial \theta}\,dx = \frac{1}{\theta-1} - \frac{(\theta-1)}{(\theta-1)^2} = 0$$

$$I(\theta) = \int_1^{+\infty} \frac{\theta-1}{x^{\theta}}\left(\frac{1}{(\theta-1)^2} - \frac{2\ln x}{\theta-1} + \ln^2 x\right)dx =$$

$$= \frac{1}{(\theta-1)^2} - \frac{2}{(\theta-1)^3} + \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2}$$

$I(\theta)$ - непр и $I(\theta)>0$ на $(1,+\infty)$ => ВМ регулярна

- $p(x,\theta) \in C^2(1,+\infty)$
- $\int_1^{+\infty} \frac{\partial^2 p(x,\theta)}{\partial\theta^2}\,dx$

$$\frac{\partial^2 p(x,\theta)}{\partial\theta^2} = \frac{\partial}{\partial\theta}\left(x^{-\theta} - (\theta-1)x^{-\theta}\ln x\right) =$$

$$= x^{-\theta}\left((\theta-1)\ln^2 x - 2\ln x\right)$$

$\int_1^{+\infty} \frac{\partial^2 p(x,\theta)}{\partial\theta^2}\,dx = 0$ => ВМ сильно регулярна

$$\sqrt{n}(\tilde{\theta}-\theta) \rightsquigarrow N\left(0, \frac{1}{I(\theta)}\right)$$

$$\sqrt{n}(\tilde{\theta}-\theta) \rightsquigarrow N(0, (\theta-1)^2)$$

$F(x_{\frac{1}{2}}) = \frac{1}{2}$

$$\int_1^{x_{1/2}} \frac{\theta-1}{x^{\theta}}\,dx = \frac{-1}{x^{\theta-1}}\Big|_1^{x_{0,5}} = \frac{1}{2}$$

$x_{0,5} = 2^{\frac{1}{\theta-1}}$

$g(\theta) = 2^{\frac{1}{\theta-1}}$

$$g'(\theta) = \ln 2 \cdot 2^{\frac{1}{\theta-1}} \cdot \left(\frac{-1}{(\theta-1)^2}\right) = -\frac{\ln 2 \cdot 2^{\frac{1}{\theta-1}}}{(\theta-1)^2}$$

$$D = \frac{[g'(\theta)]^2}{I(\theta)} = \frac{2^{\frac{2}{\theta-1}} \cdot \ln^2 2}{(\theta-1)^2}$$

$$\frac{\sqrt{n}(g(\tilde{\theta}) - g(\theta))(\tilde{\theta}-1)}{2^{\frac{1}{\theta-1}} \cdot \ln 2} \rightsquigarrow N(0,1)$$

=> асимптот. ДИ:

$$g(\tilde{\theta}) - \frac{z_{\frac{1+\beta}{2}} \ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\sqrt{n}(\tilde{\theta}-1)} < x_{1/2} < g(\tilde{\theta}) + \frac{z_{\frac{1+\beta}{2}} \ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\sqrt{n}(\tilde{\theta}-1)}$$

$\sqrt{n}(\tilde{\theta} - \theta) \rightsquigarrow N(0;1)$

$\tilde{\theta} - 1 = \frac{1}{\ln x}$

$$\sqrt{n \ln x}\,(\hat{\theta} - \theta) \sim N(0;1)$$

$$P\left(-z_{\frac{1+\beta}{2}} < \sqrt{n \ln x}\,(\hat{\theta} - \theta) < z_{\frac{1+\beta}{2}}\right)$$

$$\Updownarrow$$

$$P\left(\hat{\theta} - \frac{z_{\frac{1+\beta}{2}}}{\sqrt{n \ln x}} < \theta < \hat{\theta} + \frac{z_{\frac{1+\beta}{2}}}{\sqrt{n \ln x}}\right) \approx \beta$$